Computer Vision

Vision → ability to see

AI → machine intelligent

Machine → camera < Images, Videos } see → Intelligence

computer vision
↳ Branch of AI
↳ works with Images & videos
↳ Ability of machine to understand from Images / videos

Image
↳ what image contains ⇒ machine → pixels
↳ Group of pixels

pixel → each Dot in the Image



pixel → represents?
↳ color & shape of that portion

How pixel is represented

pixel representation format

pixel → number → 0 to 255

RGB, RGBA, Grayscale, HSV, BGR etc. --→ Channels

pixel → RGB → Channels: Red Green Blue

Red → [255 0 0] ← pixel
Green → [0 255 0]
Yellow → [255 255 0]

BGR → Blue Green Red

Blue → [255 0 0]
Red → [0 0 255]

RGBA

↳ Red Green Blue alpha

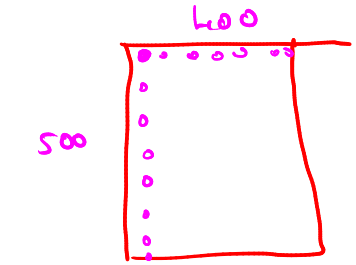
alpha → Transparency

HSV → Hue Saturation Value

Hue - color → color Range

.1.

[- - -]



Color - Image

1pixel value → 3 values

Array → numpy Array → 3 Dimension

↳ height x width x no of Channels

500 x 400 x 3

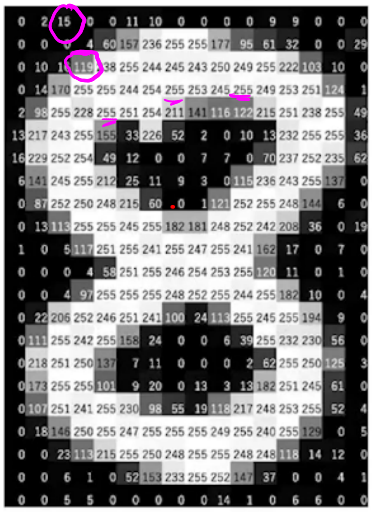
B/W Images → Gray scale

pixel → 0 to 255

0 - close → black

255 - close → white

1pixel → 1value



Gray scale

↓

numpy Array

↳ 2 Dimension

↳height x width

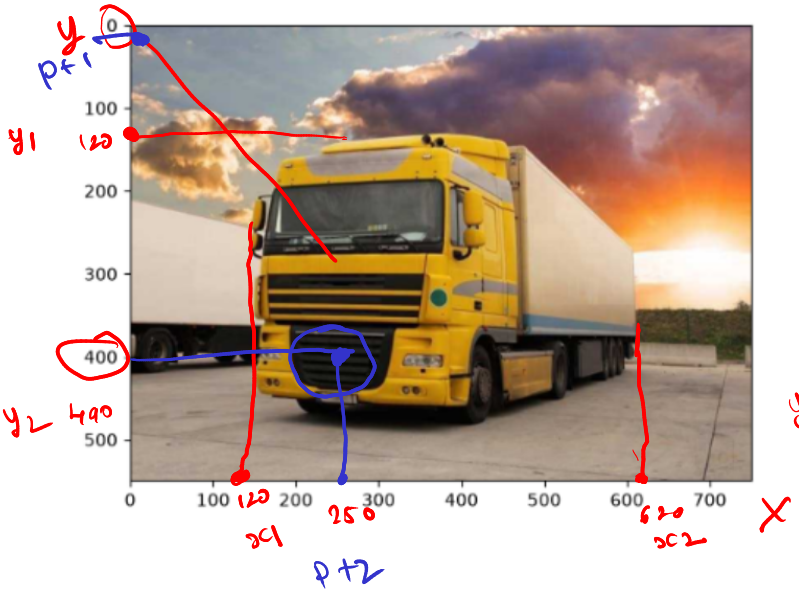
500 x 400

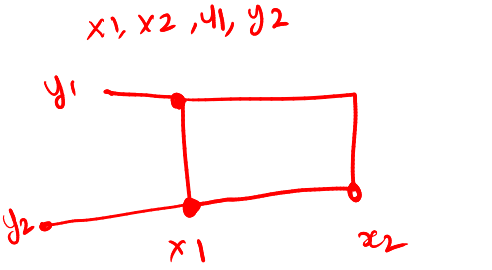
OpenCV → opensource computer vision

object detection
↳ position of the interest

why

```
5]: <matplotlib.image.AxesImage at 0x77bb5c65fd10>
```





$\frac{255+0+0}{3} = \ldots$

OpenCV

BGR to Gray

[255 0 0] ⇒ [85]

Luma transform → Luminance → weighted averaging

BGR ⇒ [20 50 30]

$\frac{20+50+30}{3} = \frac{1}{3}(20) + \frac{1}{3}(50) + \frac{1}{3}(30)$

⇒ 33.3% 33.3% 33.3%

Luma

Green = 58.7%

Red = 29.9%

Blue = 11.4%

⇒ 0.114 (20) + 0.587 (50) + 0.299 (30)

=

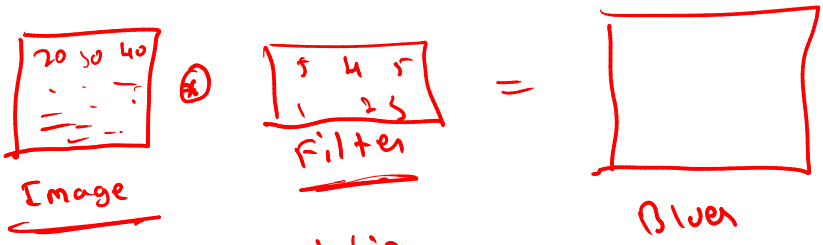
GUI ⇒ Graphical user Interface

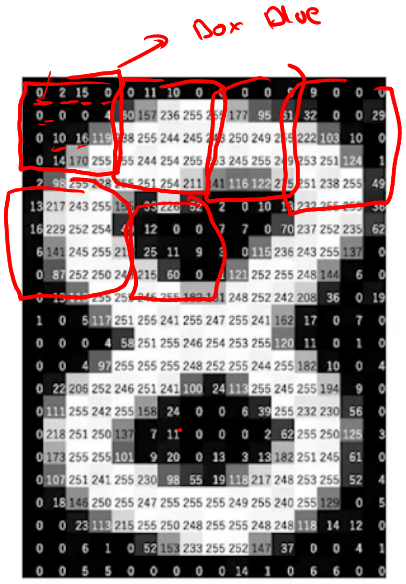
waitKey → wait & observe

Blur

↓

Filter → Blue





Box Blur

$$\frac{0+2+15+0+10+14+\cdots}{16} = 50$$

Kernal size = 4×4

Thresholding

↳ changing pixels → based on threshold

Documents - threshd

Black White	average

Binary pixel > thresh → max value

pixel < thresh → 0

Truncated

thresh = 127

pixel > thresh → thresh

pixe < thresh → pixels

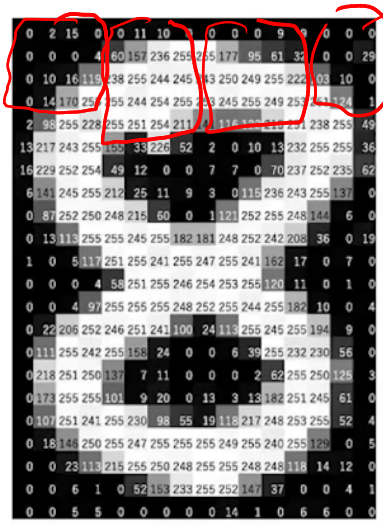
210 → 127

110 → 110

Adaptive thresholding

↳ the threshold change based region

↳ Threshold is computed regoin wise

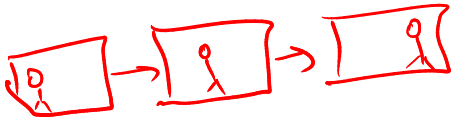


mean regoin → threshold

haussian filter → compute threshold

Video

↳ Group of Frame / seavential frames



Video

movie theater

Reels



morphological operation

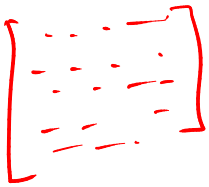
erosion

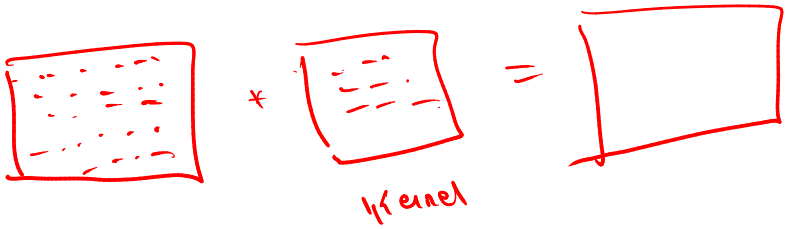
Dilation

opening

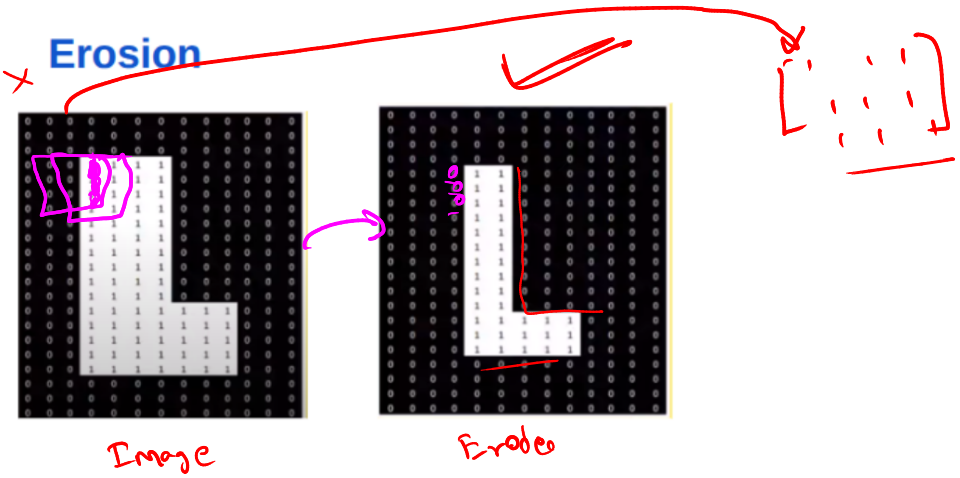
Closing

Erosion / Dilation

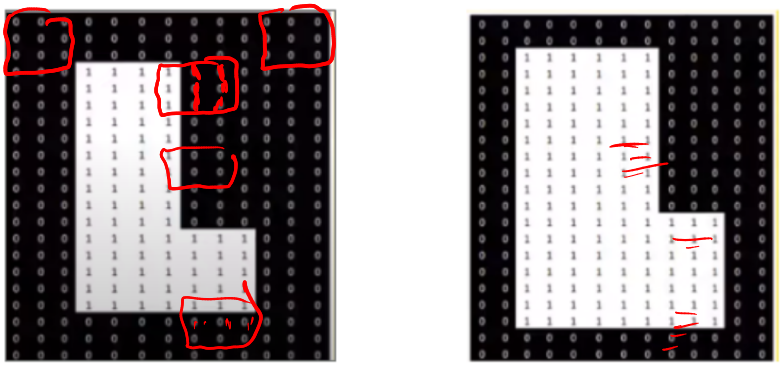




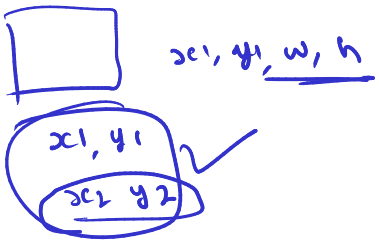
Erosion

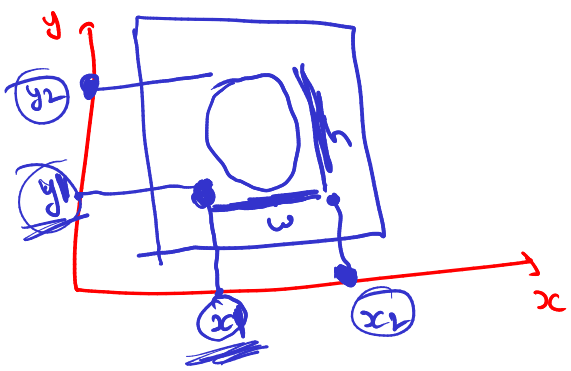


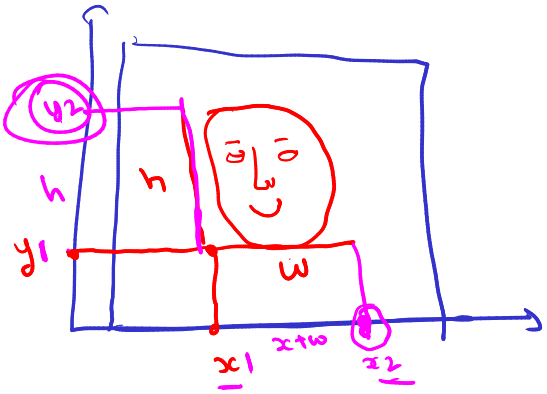
Dilation











$x_2 = x_1 + w$

$y_2 = y_1 + h$